

V FOODS Database Architecture Specification

This document details the PostgreSQL / Supabase database architecture for the V FOODS campus dining platform, including the complete schema, functions, triggers, security policies, and the Paytm three-way revenue settlement system.

1. Schema Overview & Entity Relationship



2. Table Definitions

2.1 Core Entities

outlets

Campus dining outlets and event food stalls. - `id` (text, PK): e.g. `'main-canteen'`, `'tech-cafe'`. - `name` (text, NOT NULL): Display name. - `location` (text, NOT NULL): Physical location on campus. - `is_event` (boolean, DEFAULT false): Whether outlet operates in event mode. - `is_open` (boolean, DEFAULT true): Current operating status.

menu_items

Food and drink items offered by outlets. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `outlet_id` (text, FK `outlets.id` ON DELETE CASCADE). - `name` (text, NOT NULL). - `price` (int, CHECK price > 0): In INR. - `available` (boolean, DEFAULT true). - `is_veg` (boolean, DEFAULT true). - `category` (text, DEFAULT 'General'). - `available_from` (time, NULL): Available start time window. - `available_to` (time, NULL): Available end time window. - `stock_qty` (int, NULL): Available units; NULL for unlimited. - `reserved_qty` (int, DEFAULT 0): Temporarily locked during payment pending.

profiles

User profiles extending `auth.users`. - `id` (uuid, PK FK `auth.users.id` ON DELETE CASCADE). - `full_name` (text, DEFAULT ''). - `role` (user_role ENUM: `'customer'`, `'staff'`, `'shop_admin'`, `'super_admin'`, `'admin'`). - `cust_type` (cust_type ENUM: `'student'`, `'faculty'`, `'visitor'`). - `outlet_id` (text, NULL FK `outlets.id`): Assigned stall for staff/shop admins. - `phone` (text, NULL): Used for WhatsApp/SMS notifications. - `created_at` (timestampz, DEFAULT now()).

wallets

Prepaid digital campus wallet balances. - `user_id` (uuid, PK FK `profiles.id` ON DELETE CASCADE). - `balance` (int, DEFAULT 0 CHECK balance >= 0). - `updated_at` (timestampz, DEFAULT now()).

wallet_txns

Audit ledger of all credits, debits, orders, and refunds. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `user_id` (uuid, FK `profiles.id` ON DELETE CASCADE). - `amount` (int): Positive for credits, negative for debits. - `kind` (text, CHECK `'topup'`, `'order'`, `'refund'`, `'admin_credit'`). - `ref` (text, UNIQUE): Idempotency key (e.g. `'paytm:TXN123'`, `'order:456'`). - `note` (text, NULL). - `created_at` (timestampz, DEFAULT now()).

orders

Food orders placed by students/customers. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `user_id` (uuid, FK `profiles.id` ON DELETE RESTRICT). - `outlet_id` (text, FK `outlets.id` ON DELETE RESTRICT). -

token (text, NULL): 3-digit daily counter pickup code. - **status** (order_status ENUM: 'payment_pending', 'placed', 'preparing', 'ready', 'collected', 'cancelled'). - **payment_method** (payment_method ENUM: 'wallet', 'gateway'). - **shop_payout** (int, CHECK shop_payout > 0): Raw subtotal for items. - **total** (int, CHECK total > 0): Student bill ($\text{ceil}(\text{shop_payout} * 1.05)$). - **my_profit** (int, GENERATED ALWAYS AS (shop_payout * 5 / 100) STORED). - **payment_id** (bigint, NULL FK payments.id ON DELETE SET NULL). - **pickup_slot_id** (uuid, NULL FK pickup_slots.id). - **group_id** (uuid, NULL). - **is_group_payer** (boolean, DEFAULT false). - **cancel_reason** (text, NULL). - **expires_at** (timestampz, NULL): Expiration for pending gateway orders. - **created_at** (timestampz, DEFAULT now()). - **updated_at** (timestampz, DEFAULT now()).

order_items

Line items in an order with snapshot prices. - **order_id** (bigint, FK orders.id ON DELETE CASCADE). - **item_id** (bigint, FK menu_items.id ON DELETE RESTRICT). - **name** (text, NOT NULL): Snapshot at order time. - **price** (int, NOT NULL): Snapshot at order time. - **qty** (int, NOT NULL CHECK qty > 0). - PRIMARY KEY (order_id, item_id).

2.2 Paytm Payment Architecture & Three-Way Revenue Settlement

payments

Unified ledger of all payment attempts across both Wallet Top-Up and Direct Order Payment. - **id** (bigint, GENERATED ALWAYS AS IDENTITY PK). - **user_id** (uuid, FK profiles.id ON DELETE RESTRICT). - **order_id** (bigint, NULL FK orders.id ON DELETE SET NULL). - **wallet_txn_id** (bigint, NULL FK wallet_txns.id ON DELETE SET NULL). - **amount** (numeric, CHECK amount > 0): In INR. - **payment_purpose** (text, CHECK 'ORDER', 'WALLET_TOPUP'). - **payment_method** (text, CHECK 'UPI_ID', 'UPI_APP', 'UPI_QR', 'DEBIT_CARD', 'CREDIT_CARD', 'WALLET'). - **status** (text, DEFAULT 'PENDING' CHECK 'PENDING', 'SUCCESS', 'FAILED', 'CANCELLED', 'REFUNDED'). - **paytm_order_id** (text, NULL): Merchant order reference sent to Paytm. - **paytm_txn_id** (text, NULL UNIQUE): Paytm bank transaction ID (idempotency key). - **payment_reference** (text, NULL): Internal tracking reference. - **paid_at** (timestampz, NULL): Timestamp when marked SUCCESS. - **failure_reason** (text, NULL): Paytm gateway error response message. - **created_at** (timestampz, DEFAULT now()). - **updated_at** (timestampz, DEFAULT now()).

outlet_paytm_accounts

Per-shop Paytm recipient sub-accounts with outlet-configurable split shares. - **outlet_id** (text, PK FK outlets.id ON DELETE CASCADE). - **paytm_account_id** (text, NOT NULL): Shop's registered Paytm merchant ID. - **onboarding_status** (text, DEFAULT 'PENDING' CHECK 'PENDING', 'ACTIVE', 'REJECTED'). - **settlement_enabled** (boolean, DEFAULT false). - **shop_split_percentage** (numeric, DEFAULT 90 CHECK BETWEEN 0 AND 100). - **created_at** (timestampz, DEFAULT now()). - **updated_at** (timestampz, DEFAULT now()).

platform_settlement_account

Startup platform settlement account (singleton row `id = 1`). - `id` (int, PK DEFAULT 1 CHECK id = 1). - `paytm_account_id` (text, NOT NULL): V Foods company Paytm merchant ID. - `settlement_enabled` (boolean, DEFAULT false). - `platform_split_percentage` (numeric, DEFAULT 5 CHECK BETWEEN 0 AND 100). - `created_at` (timestampz, DEFAULT now()). - `updated_at` (timestampz, DEFAULT now()).

college_settlement_accounts

Campus institution settlement accounts (one active row per campus). - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `campus_name` (text, NOT NULL): e.g. 'VIT Chennai'. - `paytm_account_id` (text, NOT NULL): University institution Paytm ID. - `onboarding_status` (text, DEFAULT 'PENDING' CHECK 'PENDING', 'ACTIVE', 'REJECTED'). - `settlement_enabled` (boolean, DEFAULT false). - `is_active` (boolean, DEFAULT true): Only one active campus at a time. - `college_split_percentage` (numeric, DEFAULT 5 CHECK BETWEEN 0 AND 100). - `created_at` (timestampz, DEFAULT now()). - `updated_at` (timestampz, DEFAULT now()). - Constraint: `CREATE UNIQUE INDEX idx_college_single_active ON college_settlement_accounts(is_active) WHERE is_active = true;`

payment_splits

Individual settlement payouts computed for every completed order. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `payment_id` (bigint, FK `payments.id` ON DELETE CASCADE). - `recipient_type` (text, CHECK 'SHOP', 'PLATFORM', 'COLLEGE'). - `outlet_paytm_account_id` (text, NULL FK `outlet_paytm_accounts.outlet_id` ON DELETE RESTRICT). - `platform_account_id` (int, NULL FK `platform_settlement_account.id` ON DELETE RESTRICT). - `college_account_id` (bigint, NULL FK `college_settlement_accounts.id` ON DELETE RESTRICT). - `split_amount` (numeric, CHECK split_amount >= 0). - `split_percentage` (numeric, NULL CHECK BETWEEN 0 AND 100). - `settlement_status` (text, DEFAULT 'PENDING' CHECK 'PENDING', 'SETTLED', 'FAILED'). - `settlement_reference` (text, NULL): Paytm disbursement transaction reference. - `settled_at` (timestampz, NULL). - `created_at` (timestampz, DEFAULT now()). - Constraint: `sql CHECK ((recipient_type = 'SHOP' AND outlet_paytm_account_id IS NOT NULL AND platform_account_id IS NULL AND college_account_id IS NULL) OR (recipient_type = 'PLATFORM' AND platform_account_id IS NOT NULL AND outlet_paytm_account_id IS NULL AND college_account_id IS NULL) OR (recipient_type = 'COLLEGE' AND college_account_id IS NOT NULL AND outlet_paytm_account_id IS NULL AND platform_account_id IS NULL))`

refunds

Audit log of processed and pending customer refunds. - `id` (bigint, GENERATED ALWAYS AS IDENTITY PK). - `payment_id` (bigint, FK `payments.id` ON DELETE RESTRICT). - `order_id` (bigint, NULL FK `orders.id` ON DELETE SET NULL). - `wallet_txn_id` (bigint, NULL FK `wallet_txns.id` ON DELETE SET NULL). - `refund_type` (text, CHECK 'ORDER_REFUND', 'WALLET_TOPUP_REFUND', 'SPLIT_REFUND'). - `amount` (numeric, CHECK amount > 0). - `status` (text, DEFAULT 'PENDING' CHECK 'PENDING', 'SUCCESS', 'FAILED'). - `paytm_refund_id` (text, NULL). - `reason` (text, NULL). - `created_at` (timestampz, DEFAULT now()). - `processed_at` (timestampz, NULL).

3. General Payment-Confirmation Flow

1. **Frontend Isolation:** The frontend client never decides or confirms a payment succeeded. It only displays a temporary "Processing" state and listens for realtime updates from Supabase.
 2. **Signed Webhook Ingestion:** When a transaction concludes, Paytm sends an HTTPS POST webhook containing `CHECKSUMHASH` and `TXNID` to the FastAPI backend at `/webhooks/paytm`.
 3. **Cryptographic Checksum Verification:** The FastAPI backend recomputes the HMAC-SHA256 signature using `PAYTM_MERCHANT_KEY`. Any unsigned, forged, or altered request is rejected with HTTP 400.
 4. **Idempotent DB Procedure:** The backend invokes `verify_and_record_payment()` in PostgreSQL. If the transaction was already processed, it returns immediately without duplicating work.
 5. **Path Execution:**
 6. For `WALLET_TOPUP`: Credits the user's wallet via `credit_wallet()` and links `wallet_txns.id`. Top-ups are never split.
 7. For `ORDER`: Marks the order `placed`, issues the 3-digit token, decrements stock, and calls `calculate_order_split()` to record the 3 split rows.
 8. **Network Reliability:** Because confirmation occurs server-to-server, if a student closes their browser or their battery dies right after paying in UPI, confirmation is unaffected.
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4. The Three-Way Revenue Split Explained

- **Shop Split (Configurable per Outlet):**
 - Each stall owner negotiates a split percentage (default 90%). Stored in `outlet_paytm_accounts.shop_split_percentage`.
 - Differing stalls can have differing rates (e.g. 90% for Main Canteen, 85% for Tech Café).
 - **Platform Split (Fixed App-Wide):**
 - V Foods platform fee (default 5%) stored in `platform_settlement_account.platform_split_percentage`.
 - **College Split (Fixed per Campus):**
 - University institutional royalty (default 5%) stored in `college_settlement_accounts.college_split_percentage`.
 - **Trigger Invariant:** A PostgreSQL trigger enforces that for any outlet: $\text{Shop \%} + \text{Platform \%} + \text{College \%} = 100\%$
 - **Penny Conservation:** The college share takes the residual difference $$(Total - Shop - Platform)$$ to guarantee exact zero-penny rounding leakage.
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5. Security & Row Level Security (RLS)

- **No Card / PIN Data:** No column stores card PAN, CVV, expiry dates, or UPI PINs.
- **RLS Access Rules:**

- **payments** : Customers view only their own payments. Staff and Shop Admins view payments for their own outlet via the linked order. Super Admins view all.
- **payment_splits** : Shop Staff/Admins can view ONLY their own outlet's **SHOP** -type rows. **PLATFORM** and **COLLEGE** rows are visible only to Super Admins. Students have zero access.
- **outlet_paytm_accounts** : Shop Admins view and manage only their outlet's row. Super Admins manage all.
- **platform_settlement_account** & **college_settlement_accounts** : Restricted exclusively to Super Admins.
- **refunds** : Users view own refunds; staff view outlet-scoped order refunds; Super Admins view all.